GAMESECURE

Technical Cybersecurity Report

Abstract

Implementation of SSH, NTP, and a site-to-site VPN using IPsec.
Threat assessment, risk analysis, and incident response plan with recommended technologies to enhance security

Zane Grose
zane.grose@hotmail.com
zane.grose@mds.ac.nz

Contents

Introduction 2

Network Design & Implementation 2

Network Diagram: 2

Configuration Steps: 3

Internet Protocol Secure Virtual Private Network (IPSec VPN): 3

Network Time Protocol (NTP): 5

Secure Shell (SSH): 6

Security Features: 6

Testing Security: 8

Security Features: 8

IPsec VPN: 10

NTP: 12

SSH: 12

Cybersecurity Threat Assessment 13

Risk Analysis 13

Social Engineering 13

Distributed Denial of Service (DDoS) Attacks 13

Cheating 13

Malware 14

Ransomware 14

Incident Response Plan 15

Recommended Security Technologies 18

Anti-Cheat 18

2FA/MFA 18

Antivirus/Antispam 18

DDoS Mitigation Tools 18

Conclusion 19

References 20

Introduction

Online gaming companies manage thousands, if not hundreds of thousands of connections per minute, consisting of sensitive data and real-time player interactions. It is imperative to reduce risks to companies sensitive data like protecting intellectual property, propriety code, and employee/user data. These gaming companies are expected by users and regulatory bodies to provide secure, reliable, and trustworthy systems, which emphasises the necessity for IPsec VPN implementation, secure shell (SSH) remote access, and timekeeping with network time protocol (NTP). Implementing IPsec VPNs provides encryption, source authentication, and provides a secure communications tunnel for communications to be transmitted.

The gaming industry is littered with threats such as DDoS, malware, cheating, social engineering, and data breach attacks. If strict standards, policies, and regulations are not followed, these attacks can easily be exploited where user accounts/systems and company networks, can be compromised. If user accounts are compromised, they can be further exploited in other areas of the internet such as social media, email, forums, and other online services. While compromised company systems can cause data breaches, loss of financial gain, and deeper malicious attacks.

Network Design & Implementation

Network Diagram:

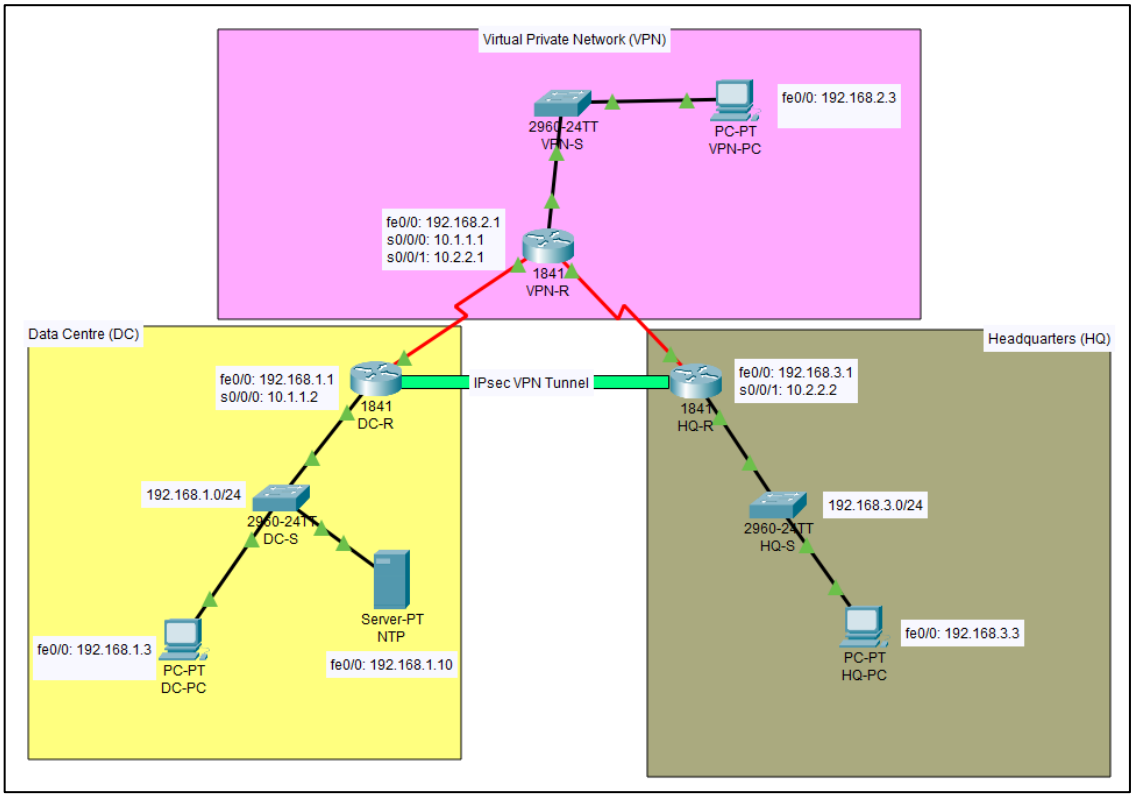


Figure 1: GameSecure Network Diagram

Configuration Steps:

Devices were setup with appropriate names and physical connections. IP's, subnets and default gateways were set for each PC to direct traffic to respective routers. Each routers IP and subnets were assigned through the physical connections, along with setting the routers to 'no shutdown', to enable routers connectivity.

Internet Protocol Secure Virtual Private Network (IPSec VPN):

1. RIP routing for each router was setup first, this was done utilizing Cisco's documentation on RIP routing (Cisco, 2024).
    a. Enabling the terminal configuration in the CLI after elevating privelieges, I enabled the RIP configuration using ***'router rip'***, then set RIP version 2 using the ***'version 2'*** command.
    b. The ***'network 192.168.X.0'*** (where X is subnet) command in the RIP configuration for the routers allowed setup of RIP routing for the IPsec VPN.
2. Traffic from the Data Centre (DC) router was marked as interesting to the Headquarters (HQ) router. This was done and confirmed using the command in *Figure 2*.

```
DC-R(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255  192.168.3.0
0.0.0.255
DC-R(config)#do show access
Extended IP access list 110
    10 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

Figure 2: Access-list 110 permitting ip traffic between routers

3. Internet Security Association & Key Management Protocol (ISAKMP) parameters were set up on policy 10. Policy 10 was accessed using the command ***'crypto isakmp policy 10'***, then the default policy configuration was noted as shown in *Figure 3*. The following steps are the setup of Phase 1 of ISAKMP.

```
DC-R(config-isakmp)#do show cryp isakmp policy

Global IKE policy
Protection suite of priority 10
        encryption algorithm:   DES - Data Encryption Standard (56 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Rivest-Shamir-Adleman Signature
        Diffie-Hellman group:   #1 (768 bit)
        lifetime:               86400 seconds, no volume limit
```

Figure 3: Default ISAKMP policy

    a. The policy's encryption was changed to AES encryption method, utilizing the command ***'encryption AES'***.
    b. Authentication method was changed to pre-share, typing the command ***'authentication pre-share'***.
    c. The Diffee-Hellman key exchange group was set to 1024-bit group, using the ***'group 2'*** command.

d. I confirm the default values for policy 10 have been changed, shown in *Figure 4*.

```
DC-R(config)#do show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit
keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #2 (1024 bit)
        lifetime:               86400 seconds, no volume limit
```

Figure 4: Configured ISAKMP policy

e. Setting the key for authentication and ensuring that authentication key (vpnpa55) is used with the correct peer address, see *Figure 5*.

```
DC-R(config)#crypto isakmp key vpnpa55 address 10.2.2.2
```

Figure 5: Command to set ISAKMP key to appropriate serial connection

4. Next was to setup Phase 2 of ISAKMP, for secure tunnel data flow.
   a. First a transform-set for use of authentication header and encryption was created using the ***'crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac'*** command.
   b. A description of the crypto map, the peer connection, the transform-set is enabled. The access control list (ACL) 110 is also linked to the crypto map, shown in *Figure 6*.

```
DC-R(config)#crypto map VPN-MAP 10 ipsec-isakmp
DC-R(config-crypto-map)#des
DC-R(config-crypto-map)#description VPN connection to HQ-R
DC-R(config-crypto-map)#set peer 10.2.2.2
DC-R(config-crypto-map)#set transform-set VPN-SET
DC-R(config-crypto-map)#match address 110
DC-R(config-crypto-map)#exit
```

Figure 6: Commands to configure the crypto map

   c. The VPN-MAP is bound to the outgoing serial interface, shown in *Figure 7*.

```
DC-R(config)#interface s0/0/0
DC-R(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

Figure 7: Commands showing the serial interface being bound to the VPN-MAP

5. Steps 2-4 were repeated for the HQ-R.
   a. *Figure 8* shows configuring site-to-site VPN with the DC-R.

```
HQ-R(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0
0.0.0.255
```

Figure 8: Access-list 110 permitting traffic between routers

   b. *Figure 9* shows configuring Phase 1 of ISAKMP.

```
HQ-R(config)#crypto isakmp policy 10
HQ-R(config-isakmp)#encryption aes
HQ-R(config-isakmp)#authentication pre-share
HQ-R(config-isakmp)#group 2
HQ-R(config-isakmp)#exit
HQ-R(config)#crypto isakmp key vpnpa55 address 10.1.1.2
```

Figure 9: Commands configuring Phase 1 of ISAKMP for HQ-R

c. *Figure 10* shows configuring Phase 2 of ISAKMP.

```
HQ-R(config)#crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac
HQ-R(config)#crypto map VPN-MAP 10 ipsec-isakmp
HQ-R(config-crypto-map)#description VPN connection to DC-R
HQ-R(config-crypto-map)#set peer 10.1.1.2
HQ-R(config-crypto-map)#set transform-set VPN-SET
HQ-R(config-crypto-map)#match address 110
HQ-R(config-crypto-map)#exit
```

Figure 10: Commands configuring Phase 2 of ISAKMP for HQ-R

d. *Figure 11* shows binding the map to the serial connection.

```
HQ-R(config)#interface s0/0/1
HQ-R(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

Figure 11: Commands configuring the serial connection to use the VPN-MAP on HQ-R

Network Time Protocol (NTP):

1. I added a physical NTP server for the Data Centre network. The NTP service was activated on this server. I changed the IP of the NTP server to *'192.168.1.10'*, then enabled authentication for the service, shown in *Figure 12*

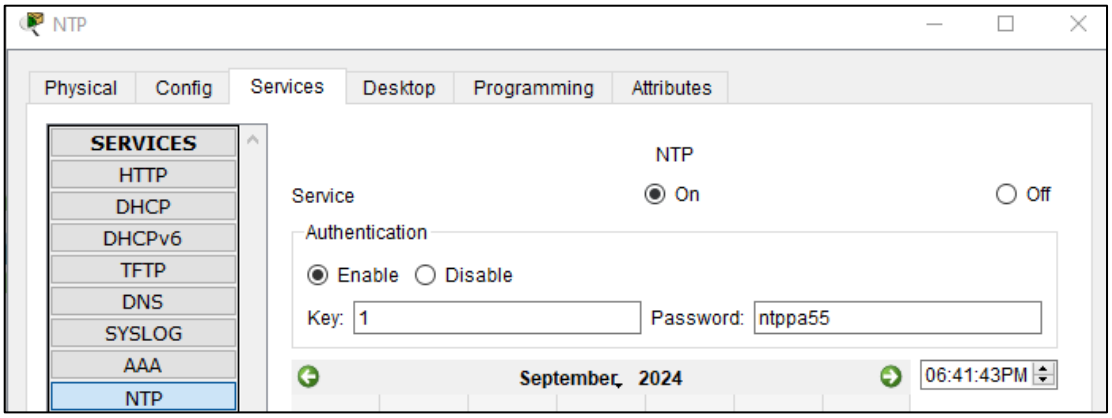


Figure 12: GUI for NTP server, showing authentication being enabled

2. I then accessed the DC-R CLI and checked the status of NTP and the current clock time to confirm NTP was disabled - acting as a baseline, see *Figure 13 & 14*

```
DC-R#sh clock detail
*4:17:23.664 UTC Mon Mar 1 1993
Time source is hardware calendar
```

Figure 14: Output displaying default hardware clock

```
DC-R#sh ntp status
%NTP is not enabled.
```

Figure 13: Output displaying NTP is disabled

3. Confirming NTP was not active, I entered the configuration of the terminal on DC-R using ***'config terminal'*** and utilized the ***'ntp server 192.168.1.10'*** to set the NTP server as the time-source.
4. I then finished configuration of the NTP by authenticating time sources, referencing the trusted-key, and entering the password set for the NTP server service, shown in *Figure 15*.

```
DC-R(config)#ntp authenticate
DC-R(config)#ntp trusted-key 1
DC-R(config)#ntp authentication-key 1 md5 ntppa55
```

Figure 15: Authenticating time-source, referencing trusted-key, and confirming time-source connection with password

5. Steps 1-4 were repeated for HQ-R and VPN-R, to synchronize times across each network.

Secure Shell (SSH):

SSH was only enabled for routers, all switches VTY lines were disabled.

1. SSH was first enabled on the Data Centre router (DC-R). This begun with setting the domain name and the hostname. Hostname was set using the ***'hostname DC-R'*** command, and the domain name using ***'ip domain-name datacentre'*** in the configuration terminal for the router.
2. I then generated encryption keys, utilizing the ***'crypto key generate rsa'*** command, using 1024 bits in the modulus.
3. A local login was needed next, this was done using the command ***'username dcadmin password dcadminpa55'*** .
4. The version of SSH is specified using ***'ip ssh version 2'***.
5. The virtual teletype (VTY) lines were configured to use SSH, utilizing the following commands:
    a. ***'line vty 0 4'***
    b. ***'transport input ssh'***
    c. ***'login local'***
6. Steps 1-5c were repeated for VPN-R and HQ-R. The domain, local username and passwords are different for each router.
    a. Relevant passwords for all devices and console lines are listed in the 'GameSecure Passwords.txt' file submitted alongside this report.

Security Features:

All steps regarding the console and EXEC mode access are accessed using set passwords, listed in the 'GameSecure Passwords.txt' file during configuration. These are only listed in the .txt file for marking and testing purposes. In a live situation, these would be stored on a piece of paper or secure password storage application, only accessible by authorized employees.

Added a password to access console of all routers and switches:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Accessed terminal configuration using ***'config terminal'***
3. Accessed console line using ***'line console 0'***
4. Set the password using the ***'password conpa55'*** command

Added a secret to access EXEC mode of the consoles:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Accessed terminal configuration using ***'config terminal'***
3. Set the enable secret using the ***'enable secret enpa55'*** , where 'enpa55' is the relative passwords listed in the 'GameSecure Passwords.txt' document

Added a MOTD banner to notify users only authorized users are allowed into the console:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Accessed terminal configuration using ***'config terminal'***
3. Set MOTD banner using ***'banner motd '!--- GameSecure authorized access only! Turn back if un-authorized! ---!"*** command on each router/switch.

Saved NVRAM so reset settings are secure:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Used the ***'copy running-config startup-config'*** command to copy the secure configuration to the NVRAM.

All VTY lines on Switches have login disabled, transport input set to none, and NVRAM (startup-config) saved from the running-config:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Accessed terminal configuration using ***'config terminal'***
3. Accessed all VTY lines using ***'line vty 0 15'***
4. Disabled logins using the ***'no login'*** command
5. Set the transport input to none using ***'transport input none'***
6. Exited the VTY line config and terminal config by typing ***'exit'*** twice.
7. In EXEC console mode, copied the running config to the startup using ***copy running-config startup-config'***

Ensured all plain-text passwords were encrypted:

1. Accessed the EXEC privileges using ***'enable'*** in the router/switch CLI
2. Accessed terminal configuration using ***'config terminal'***
3. Encrypted passwords using ***'service password-encryption'*** command

Testing Security:

Security Features:

All security feature was tested on each router and switch. I'll be using the Data Centre router and switch to show the testing method in this section, but testing was done on all routers & switches.

Console Passwords & MOTD banner:

The CLI for the router was accessed, prompting me for a password. *Figure 16* confirms that console password is setup correctly, the MOTD banner is also shown.

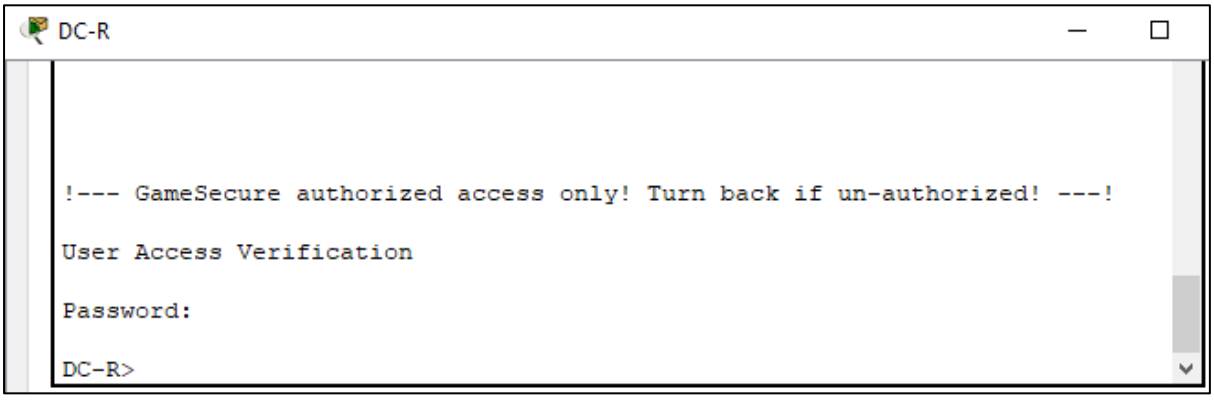


Figure 16: MOTD Banner & confirmation that console is secure

Console EXEC mode:

After accessing the console by entering the console password, the ***'enable'*** command was entered, proving that the enable secret has successfully been activated, shown in *Figure 17*.

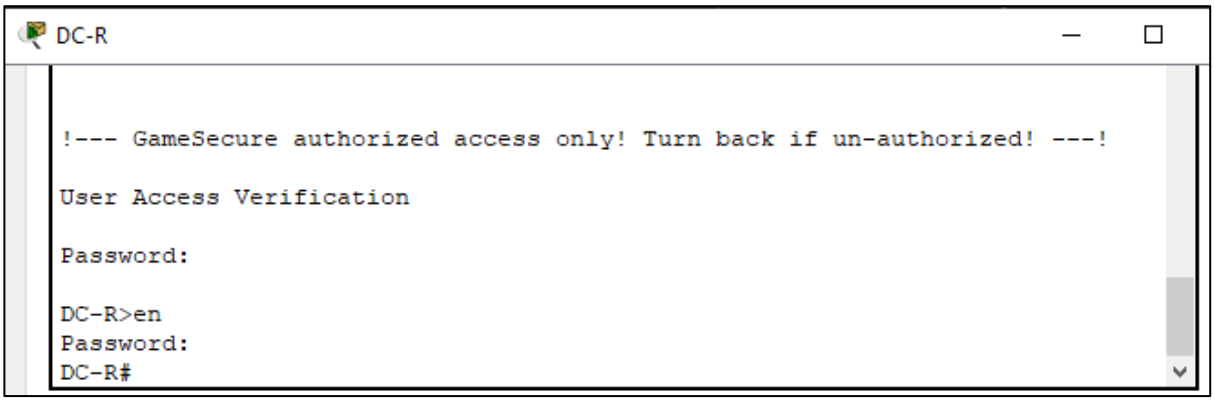


Figure 17: Confirmation that 'enable' command is secure behind a secret (password)

NVRAM configuration & plain-text password encryption:

Saved NVRAM configuration is shown in *Figure 18-20* show that all plain-text passwords (such as console line 0) have been encrypted, and that the edited running-config has successfully been saved to the NVRAM.

```
crypto isakmp policy 10
 encr aes
 authentication pre-share
 group 2
!
crypto isakmp key vpnpa55 address 10.2.2.2
!
!
!
crypto ipsec transform-set VPN-SET esp-3des esp-sha-hmac
!
crypto map VPN-MAP 10 ipsec-isakmp
 description VPN connection to HQ-R
 set peer 10.2.2.2
 set transform-set VPN-SET
 match address 110
!
!
!
!
ip ssh version 2
ip domain-name datacentre
!
!
spanning-tree mode pvst
```

Figure 19: Startup-config settings, showing ISAKMP policy, plain-text ISAKMP key, VPN-MAP, and SSH settings

```
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/0/0
 ip address 10.1.1.2 255.255.255.252
 clock rate 2000000
 crypto map VPN-MAP
!
interface Serial0/0/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router rip
 version 2
 network 10.0.0.0
 network 192.168.1.0
 network 192.168.2.0
 network 192.168.3.0
 no auto-summary
```

Figure 18: Startup-config settings, showing saved connection information and RIP settings

```
DC-R#show startup
Using 1676 bytes
!
version 12.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname DC-R
!
!
!
enable secret 5 $1$mERr$TYyS0iHX8M5axkfDSNXsb0
enable password 7 0824425E084C50
!
!
!
!
!
!
no ip cef
no ipv6 cef
!
!
!
username dcadmin password 7 08254F4F0D140C19020A5951
```

Figure 20: Startup-config settings, showing encrypted passwords

Switches VTY lines and NVRAM:

Figure 21 shows that the NVRAM for the DC-S switch has been saved from the running-config, disabling the VTY lines.

```
interface Vlan1
 no ip address
 shutdown
!
banner motd ^C!--- GameSecure authorized access only! Turn back if un-
authorized! ---!^C
!
!
!
line con 0
 password 7 0822434019185042
 login
!
line vty 0 4
 no login
 transport input none
line vty 5 15
 no login
 transport input none
```

Figure 21: Startup-config (NVRAM) of DC-S, showing VTY lines have been disabled

IPsec VPN:

1. To test the IPsec VPN is working as expected, I initially showed the IPsec security associations (SA) information with the command *'show crypto ipsec sa' (*See *Figure 22)*. This is the baseline for the packets that are encapsulated, encrypted, decapsulated, and decrypted.

```
DC-R#show crypto ipsec sa

interface: Serial0/0/0
    Crypto map tag: VPN-MAP, local addr 10.1.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
     current outbound spi: 0x0(0)
```

Figure 22: IPSec Security Associations information, before any traffic has been sent through the IPsec VPN tunnel

2. I then used the *'ping 192.168.3.3'* (HQ-PC) command to send 'interesting traffic' through the IPsec VPN tunnel (see *Figure 23*).

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.3.3

Pinging 192.168.3.3 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.3.3: bytes=32 time=7ms TTL=126
Reply from 192.168.3.3: bytes=32 time=14ms TTL=126

Ping statistics for 192.168.3.3:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 14ms, Average = 10ms
```

Figure 23: Pinging the Data Centre PC (DC-PC) from the Headquarters PC (HQ-PC), to send 'interesting traffic' through the VPN tunnel.

3. Reevaluating the IPsec SA information, I can see that packets from DC-PC to HQ-PC have been encrypted, encapsulated, decapsulated, and decrypted, shown in *Figure 24*.

```
DC-R#show crypto ipsec sa

interface: Serial0/0/0
    Crypto map tag: VPN-MAP, local addr 10.1.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
   #pkts decaps: 2, #pkts decrypt: 2, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
     current outbound spi: 0xFF1E0027(4280156199)
```

Figure 25: IPsec Security Association information after sending 'interesting traffic' through the VPN tunnel

4. To ensure the IPsec VPN is only utilizing 'interesting traffic', I created 'uninteresting traffic' by pinging from DC-PC to VPN-PC (see *Figure 25*).

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time=3ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=10ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 4ms
```

Figure 24: Pinging the VPN-PC from the Data Centre PC to create 'uninteresting traffic' through the VPN tunnel

5. Finally, I check the IPsec SA information again, confirming that 'uninteresting traffic' is not encrypted (see *Figure 26*) and that the SA information has not changed.

```
DC-R#show crypto ipsec sa

interface: Serial0/0/0
    Crypto map tag: VPN-MAP, local addr 10.1.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.3.0/255.255.255.0/0/0)
   current_peer 10.2.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 3, #pkts encrypt: 3, #pkts digest: 0
   #pkts decaps: 2, #pkts decrypt: 2, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 10.1.1.2, remote crypto endpt.:10.2.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/0/0
     current outbound spi: 0xFF1E0027(4280156199)
```

Figure 26: IPsec SA information after sending 'uninteresting traffic' from the DC-PC to the VPN-PC

NTP:

1. To confirm the NTP server was setup correctly, I took the information from step 2 of 'NTP configuration' and referenced it against the same commands after the configuration was complete.
2. First was to see if the NTP service was setup on the DC-R correctly, this was done using the command ***'show ntp status'***, shown in *Figure 27*

```
DC-R#show ntp status
Clock is synchronized, stratum 16, reference is 192.168.1.10
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is 25D8B22D.000000C1 (7:47:57.193 UTC Sun Apr 30 2056)
clock offset is 0.00 msec, root delay is 1.00  msec
root dispersion is 249.39 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193
s/s system poll interval is 4, last update was 12 sec ago.
```

Figure 27: NTP status information on DC-R

3. Finally, confirming by checking the clock detail on the router, see *Figure 28*.

```
DC-R#show clock detail
18:53:1.628 UTC Sun Sep 29 2024
Time source is NTP
```

Figure 28: Clock detail on DC-R, utilizing NTP

4. Steps 1-3 were repeated for VPN-R and HQ-R, to confirm synchronized time-sources on each router.

SSH:

1. Testing the SSH was done using the DC-PC. I opened the Command Prompt interface of the PC and entered the command ***'ssh -l dcadmin 192.168.1.1'.***
2. The password prompt is requested, to which I enter the password ***'dcadminpa55'.***
3. The router is accessed through the DC-PC, shown in *Fig*ure 29.

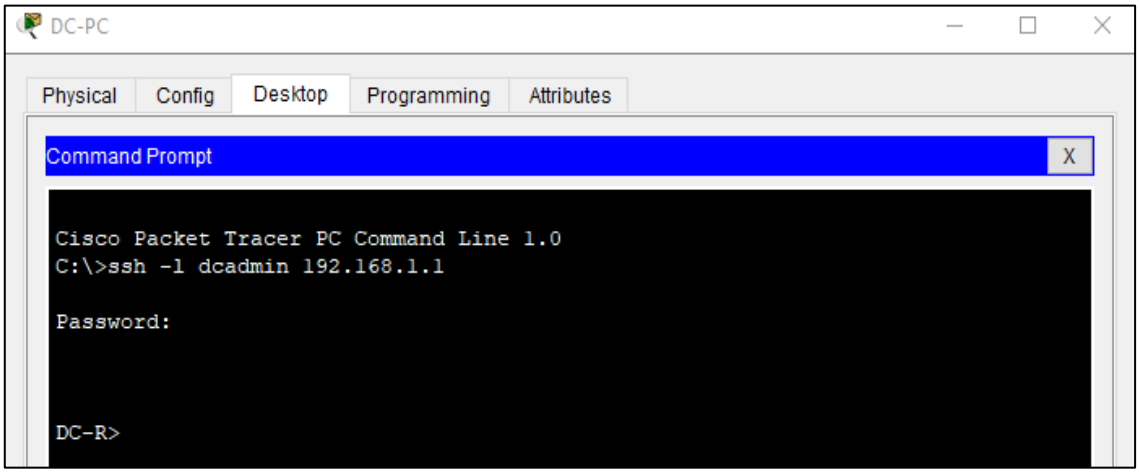


Figure 29: DC-R console accessed from the DC-PC using SSH local login

4. Steps 1-3 were repeated for each router (VPN-R, HQ-R) from DC-PC, using the respective SSH login details outlined in the 'GameSecure Passwords.txt' file.

Cybersecurity Threat Assessment

Risk Analysis

Social Engineering

One attack that plagues the gaming industry today are social engineering attacks. One such attack could consist of an unaware user engaging with an attacker over a game, over text chat, voice chat, forums, or third-party platforms. An attacker aims to gain trust, attempting to manipulate the user into visiting malicious links, websites, or trade in-game items on basis of trust.

Another form of social engineering is creating fake emails posing as a legitimate company entity, such as emailing the user that their account is compromised and needs to be reviewed to get the users immediate attention. If the user is tricked, the user could download malicious programs to their computer or enter their personal information into compromised websites. This can lead to users accounts being compromised, or user systems being used in botnets - further being used to attack the services or systems using DDoS attacks.

Distributed Denial of Service (DDoS) Attacks

DDoS attacks aim at halting the availability of a service, and usually done by attackers using infected computers which are connected, creating a botnet. Botnets are a network of infected computers and aim to send an overwhelming amount of fake traffic to a service or system, so the system/service crashes or becomes slow, effectively hindering availability for. These attacks are typically performed by people who want to push an agenda, looking for financial gain, or disgruntled individuals. DDoS attacks can cause financial loss for the company due to downtime, decline of trust from users, and potential reputational damage, which can cause long-term business impacts.

Cheating

Cheating has been an issue for a long time in the gaming industry, this type of attack aims at finding vulnerabilities in the software code, allowing attackers to write their own software to gain advantages in the game. Some examples include providing themselves with extra in-game resources, automatic aiming, God-like powers (no death, teleporting), and lag switching (delaying network traffic), most – if not all – of these cheats offer the attacker with an advantage over others in the game. Cheats can severely hinder a company's trustworthiness, the experience of legitimate users, and compromise software code. Cheats can lead to further vulnerabilities in the system, as vulnerable code can lead to further exploitation, possibly accessing sensitive data, such as company intellectual information or user/employee information. Cheating can also lead

to financial loss through player mistrust, and in turn, refunding the game, or spreading the word about the game not being reliable, preventing new users from trying the game.

Malware

Malware is a nasty software that can affect data, systems, and services. Malware can be installed on a user's computer by utilizing social engineering attacks, visiting compromised websites, or clicking on malicious ads/links. There are many types of malwares such as worms that self-replicate automatically through and infected system, viruses that attach to legitimate files and spread when they are shared, trojans that disguise themselves as legitimate software used to gain unauthorized access, and spyware that monitors users activity. Malware can cause data breaches, compromise systems, financial loss, and user mistrust.

Ransomware

Like malware, ransomware can be installed through infectious attachments, malicious links, or infected software downloads. One type of ransomware locks users out of company systems, demanding a ransom. Another is like the first described attack, but instead of just locking systems, it encrypts the data and demands a ransom for the decryption key. There is also scareware, a form of ransomware that prompts a user to download a software or pay ransoms by abusing emotional responses from authorized users. These attackers aimed at getting financial gains by holding data and systems hostage. These are some of the most common attacks, aimed at medium to large companies due to the potential monetary gain.

Incident Response Plan

The following table is a baseline documentation on how to handle cyber-attacks listed in the 'Risk Analysis' section in 4 steps.

CYBERATTACK	CONTAINMENT	ERADICATION	RECOVERY	POST-ATTACK
SOCIAL ENGINEERING (PHISHING, SCAM)	Identify the employee/user who has been manipulated and determine what type of social engineering attack has occurred. Restrict account control of the compromised employee/user. If malware has been installed, contain the employees system from other systems or services. Notify relevant teams of the attack.	Identify the medium of which the attack happened, ban those malicious accounts (such as email, IP's, social media). If malware has been installed, identify the malicious software and remove it ASAP.	Restore systems to previous state if possible. Prompt employee/user to change their account details. Restrict access on server/services	Educate users on common social engineering attacks to prevent future mishaps. Alert management on the attack, emphasising the importance on educating users/employees.
DDOS	Determine the IP(s) sending the DDoS attack using logs, intrusion detection systems or network monitoring tools, restrict traffic from that IP(s) using ACLs. If this is ineffective, attempt to change the router IP, server IP, or website URL's to buy	Ban and/or restrict the attackers IP(s) and possible botnet IPs and restrict traffic from them. Rate limit the system, limiting the number of requests users can perform at one time. Implement filtering	Restore throttled or shutdown servers/services. Review the network infrastructure to check for system vulnerabilities or shortcomings. Communicate with stakeholders and managers	Analyse the DDoS attack and what layer the attack focused on, then update protocols on that layer. Revisit, analyse, and implement protection methods, such as firewalls, proxies, load-balancers, and scrubbing servers.

	time. If this is ineffective, the affected service(s) may have to be shut down temporarily, as a last resort.	technologies to filter attacks before reaching the service.		
CHEATING	Ban all user accounts suspected of cheating to minimize the widespread use of the cheat.	Analyse the cheat used and identify the vulnerable software code, patch as soon as possible after testing thoroughly, as to not introduce more vulnerable code.	If other users are affected by gaining or losing items or resources in game, attempt to restore to an earlier stage.	Perform a review of the cheat and how it occurred. Monitor the game for the cheat and similar cheats to avoid another incident. Communicate with the community, assuring the threat has been contained.
MALWARE	Isolate the affected system(s) from the wider network. Analyse logs to see where the source of malware came from and isolate that system from the network and internet.	Remove malware using updated anti-virus/malware software. Patch any vulnerabilities that were exploited in the attack.	Restore systems to before the attack, as back-doors could have been introduced to the systems. Monitor affected systems for any residual malware or unusual activity.	Conduct a review of the attack to understand how it happened and how to respond in the future to a similar attack. Educate users on recognizing threats. Remove access from unsafe websites, such as websites that don't support HTTPS
RANSOMWARE	Isolate affected system(s) from the network and internet. Determine what files have been locked/encrypted	Identify the type of ransomware used. Use anti-virus/malware software to	Restore systems to a state before ransomware was detected using backups. Monitor the affected	Review security policies to identify weaknesses in security posture. Revise and update security

	to assess impact.	remove (if possible) the ransomware.	systems and data after restoration to identify any signs of reinfection.	plans to enhance response against future attacks. Educate users on ransomware threats, safe practices, and how to respond in event of an attack.

Recommended Security Technologies

Anti-Cheat

Anti-cheat software such as Easy Anti-Cheat (EAC), BattlEye, Valve Anti-Cheat (VAC), and SARD Anti-Cheat are reputable software that are widely used in today's gaming industry. This type of software aims at detecting unusual user gameplay by using algorithms and in some newer cases, machine learning and artificial intelligence.

2FA/MFA

Two-Factor Authentication (2FA) or Multi-Factor Authentication (MFA) can help secure users against account compromising attacks. This prevents attackers from directly accessing a user's account by only knowing a user's username and password. Typical forms of 2FA/MFA consist of SMS codes, authenticator applications, and biometric data, providing a client-side method of preventing account hacks.

Antivirus/Antispam

Antivirus or Antispam software is aimed at detecting malicious software, generating alerts, and preventing malware from infecting hosts. This type of software provides critical protection for important systems with another layer of security against commonly known malware, viruses, and spam attacks.

DDoS Mitigation Tools

Companies such as Cloudflare, AWS, Microsoft, and Radware provide DDoS mitigation tools. This software mitigates the effects of DDoS on a company system by routing traffic to external third-party servers, keeping the servers up and handling the attack. There are two types of mitigation software (Cloudfare, 2022).

- On-demand software allows normal traffic flow, until a DDoS attack is identified, where mitigation methods are implemented.
- Always-on software is always monitoring network traffic, ensuring only clean traffic reaches systems. Ensuring 24/7 availability of the service.

Security Information and Event Management (SIEM)

SIEM systems allow IT employees to analyse logs from multiple sources on the network in an organized way, allowing for quick response in event of an attack. A SIEM allows for incident monitoring and response, event correlation, and log management, which assists in detecting suspicious user activity, monitoring user behaviour, limiting access/requests, and generating compliance reports (Microsoft, 2024)

Conclusion

The network I have setup incorporates fundamental security features, including password-protected access to console lines and console EXEC modes, along with a MOTD banner informing users to the authorized nature of the devices. Saving the edited configuration to the NVMRAM ensures reboots are safe from manipulation. Disabling VTY lines on Switches mitigates the risk of unauthorized access and encrypting all passwords to avoid over-the-shoulder attacks on authorized users accessing sensitive information.

Along with these security features, I've added a SSH method to access all routers, allowing authorized users to securely access routers remotely. Configuration of NTP for each router ensures synchronization of time across networks, allowing for accurate log-keeping and auditing. Including the IPsec VPN tunnel established between the Data centre and Headquarters allows for secure communications, encrypting sensitive communications and authenticating sources, ensuring the integrity of sensitive information.

The cybersecurity threat assessment contains a risk analysis of common threats plaguing the gaming industry today, from malware to social engineering attacks. The risk analysis is complimented by an incident response plan for each of these attacks, detailing a containment, eradication, recovery, and post-attack actions, which aims to guide the incident response team in event of a compromising attack.

Lastly, I suggested to implement some software such as anti-cheat and anti-virus/spam, which can avoid malware and spam attacks, and prevent cheats from being used. 2FA/MFA which helps in avoiding compromising user accounts, even when passwords are compromised. DDoS mitigation tools help mitigate the effects of DDoS attacks on systems and services, by diverting traffic to third-party servers and automatically filtering malicious traffic before reaching critical systems. Finally, suggesting a SIEM system will allow centralized log monitoring, easy network traffic tracking, and allow proactive responses to real-time attacks.

References

Cisco. (2024, 09). *Configuring RIP*. Retrieved from Cisco: https://www.cisco.com/c/en/us/td/docs/switches/lan/catalyst9400/software/release/16-9/configuration_guide/rtng/b_169_rtng_9400_cg/configuring_rip.pdf

Cloudfare. (2022). *Five best practices for mitigating DDoS attacks*. Retrieved from Cloudfare Assets: https://cf-assets.www.cloudflare.com/slt3lc6tev37/58Znmio29pRXDLKoQgNIz4/5cf1a6d3b1b1f5f1ea995460e04eb512/BDES-2587-Design-Wrap-Refreshed-DDoS-White-Paper-Letter.pdf?_gl=1*1m6y5ee*_gcl_au*OTczNzc2MjczLjE3MjUxNjAzNTU.*_ga*YTY5MDgzOWItODI0Ny00ZjE2LWExY2EtNj

Microsoft. (2024). *SIEM Defined*. Retrieved from Microsoft Security: https://www.microsoft.com/en-nz/security/business/security-101/what-is-siem